SolidWorks Project

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Introduction

In this project, I have designed a bending roller machine using SolidWorks. This machine is mainly used to bend sheet metal by passing it between rollers. The purpose of this work was to create the 3D parts, assemble them, understand the mechanism, and simulate the working of the machine.

Theory

A bending roller machine works on the principle of plastic deformation. When a metal sheet passes between rollers, it experiences both tensile and compressive stresses. These stresses bend the sheet into a curved shape. The neutral axis remains the same length, while the outer surface stretches and the inner surface compresses. Factors like sheet thickness, yield strength, and roller spacing affect the bending.

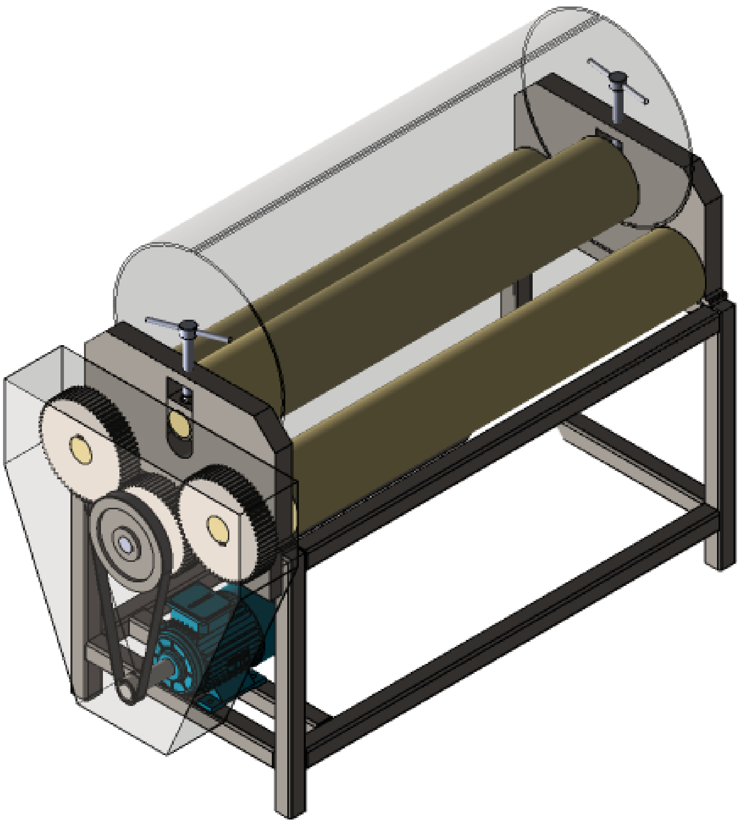

Figure 1: SolidWorks Model of the Bending Roller Machine

Working Principle

The bending roller machine usually consists of three rollers—two at the bottom and one on the top. The sheet is placed between these rollers. By adjusting the position of the top roller, the bend radius can be controlled. When the rollers rotate, the sheet moves forward and gradually bends. In SolidWorks, each component such as the frame, rollers, shafts, and supports was designed and assembled. Motion simulation was used to show how the sheet bends as it passes through the rollers.

Application

Bending roller machines are commonly used in industries that involve sheet metal fabrication. Some applications include:

- Manufacturing pipes, tubes, and cylindrical shapes
- Forming curved metal panels for construction
- Making parts for automobiles and machinery
- Producing storage tanks, drums, and metal shells

Conclusion

The SolidWorks project helped in understanding the design of a bending roller machine and how its parts work together. By creating a 3D model and simulating motion, the working principle became much clearer. This project also improved knowledge about assembly, constraints, and sheet metal bending concepts.

Reference

YouTube Video Link Used : `https://youtu.be/8OaWzXoOUFE?si=YryFniJDcyFBGnSb`